

A Jump Obstruction to Quadratic-Variation-Only Estimates

Run `fa.banach_001`, lane 7

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Source and Question

The source paper is Mark Veraar and Ivan Yaroslavtsev, *Cylindrical continuous martingales and stochastic integration in infinite dimensions*, arXiv:1602.03996. At the end of the introduction the authors ask what structure of a cylindrical *noncontinuous* local martingale is required in order to build a stochastic integration theory with two-sided estimates for stochastic integrals.

In the continuous theory of the source paper, the main size datum is the cylindrical quadratic variation $[[M]]$, together with its operator derivative. The following elementary example shows that, after jumps are allowed, a theory whose estimates see only the predictable second-order quadratic characteristic cannot work. Thus extra jump-size structure is not cosmetic; it is forced already in the scalar case.

Idea of the Proof

A rare jump of size R and intensity R^{-2} has unit predictable quadratic variation. Hence all these martingales look identical to any norm that records only the second characteristic. But for $p > 2$, the p -th moment is dominated by the event that the rare jump occurs once, giving size approximately R^p with probability approximately R^{-2} . The resulting moment grows like R^{p-2} .

Theorem 1. *Let $p > 2$. There is no constant C_p with the following property: for every real-valued purely discontinuous square-integrable martingale M on $[0, 1]$ with predictable quadratic variation $\langle M \rangle_t = t$,*

$$(\mathbb{E}|M_1|^p)^{1/p} \leq C_p.$$

Consequently, no noncontinuous cylindrical stochastic integration theory can have uniform two-sided L^p estimates for the constant integrand if the right-hand side depends only on the predictable quadratic variation/intensity measure of the scalar projections.

Proof. Let $R \geq 2$, and let $N^R = (N_t^R)_{0 \leq t \leq 1}$ be a Poisson process with intensity $\lambda_R = R^{-2}$. Define

$$M_t^R = R(N_t^R - \lambda_R t), \quad 0 \leq t \leq 1.$$

Then M^R is a real-valued square-integrable martingale with jumps of size R . Since the predictable quadratic variation of the compensated Poisson martingale $N_t^R - \lambda_R t$ is $\lambda_R t$, we have

$$\langle M^R \rangle_t = R^2 \lambda_R t = t, \quad 0 \leq t \leq 1.$$

Thus all martingales M^R have the same predictable second characteristic.

On the event $\{N_1^R = 1\}$,

$$|M_1^R| = R|1 - \lambda_R| \geq R/2$$

for $R \geq 2$. Also

$$\mathbb{P}(N_1^R = 1) = \lambda_R e^{-\lambda_R} = R^{-2} e^{-R^{-2}} \geq e^{-1/4} R^{-2}.$$

Therefore

$$\mathbb{E}|M_1^R|^p \geq (R/2)^p e^{-1/4} R^{-2} = 2^{-p} e^{-1/4} R^{p-2}.$$

This tends to infinity as $R \rightarrow \infty$, contradicting any bound depending only on p and on the common value $\langle M^R \rangle_1 = 1$.

The same example is a one-dimensional cylindrical martingale, so it embeds unchanged into any broader cylindrical framework. For the stochastic integral of the constant scalar integrand 1, the integral is just M^R , so the failure is already a failure of the desired integral estimate. $\square$

Remark 1 (What the obstruction does not say). *This does not contradict Burkholder–Davis–Gundy inequalities for martingales with jumps. Those inequalities use richer data, for instance the optional quadratic variation $[M^R]_1 = R^2 N_1^R$, whose $L^{p/2}$ -size also grows with R . Nor does it contradict jump integration theories with Rosenthal-type or Poissonian norms. The point is only that the continuous paper’s quadratic-variation-only philosophy cannot be transplanted to jumps using only predictable second characteristics.*

Verification Notes

The example is scalar, so no Banach-space geometry is hidden. The predictable quadratic variation computation is the standard one for compensated Poisson processes. The lower bound uses only the one-jump event and is uniform for $R \geq 2$.

Novelty and Literature Check

The run indexes were searched for arXiv:1602.03996, the source title, “cylindrical noncontinuous”, “two-sided estimates”, and “scalar quadratic variation”. No existing run packet for this source was found. A bounded web search on 2026-06-21 for exact phrases around the source’s $[[M]]$ question and tensor/scalar quadratic variation did not find a matching packet-level answer. The phenomenon is elementary and should be regarded as a folklore-level obstruction unless a deeper novelty check says otherwise.

References

- M. Veraar and I. Yaroslavl'tsev, *Cylindrical continuous martingales and stochastic integration in infinite dimensions*, Electron. J. Probab. 21 (2016), no. 59, 1–53; arXiv:1602.03996.
- S. Dirksen, *Itô isomorphisms for L^p -valued Poisson stochastic integrals*, Ann. Probab. 42 (2014), no. 6, 2595–2643. Cited for context on richer Poisson/Rosenthal jump norms, not used in the proof.

Human-Review Recommendation

Verify the scope: this packet proves a negative result for quadratic/intensity-only jump estimates. It should not be read as a negative answer to the full broad program of constructing noncontinuous cylindrical integration theories with additional jump structure.